

Antimicrobial Traits of Tea- and Cranberry-Derived Polyphenols against Streptococcus mutans

There are over 750 species of bacteria that inhabit the human oral cavity, but only a small fraction of those are attributed to causing plaque-related diseases such as caries. *Streptococcus mutans* is accepted as the main cariogenic agent and there is substantial knowledge regarding the specific virulence factors that render the organism a pathogen. There has been rising interest in alternative, target-specific treatment options as opposed to nonspecific mechanical plaque removal or application of broad-spectrum antibacterials that are currently in use. The impact of diet on oral health is undeniable, and this is directly observable in populations that consume high quantities of polyphenol-rich foods or beverages. Such populations have low caries incidence and better overall oral health. *Camellia sinensis*, the plant from which various forms of tea are derived, and *Vaccinium macrocarpon* (American cranberry fruit) have received notable attention both for their prevalence in the human diet as well as for their unique composition of polyphenols. The biologically active constituents of these plants have demonstrated potent enzyme-inhibitory properties without being bactericidal, a key quality that is important in developing therapies that will not cause microorganisms to develop resistance. The aim of this review is to consider studies that have investigated the feasibility of tea, cranberry, and other select plant derivatives as a potential basis for alternative therapeutic agents against *Streptococcus mutans* and to evaluate their current and future clinical relevance.